Automated Photopolymer Micro-Dispenser

Technical Design Report

The Dalton Lab – Microneedle Research Group

Abstract

This report describes the development of a programmable automated fluid dispenser capable of precisely dispensing small volumes of non-Newtonian photopolymers onto solid microneedle arrays. The system enables controlled deposition of photopolymer insulation on microneedle bases, allowing for precise control over insulation height. A prototype was constructed and successfully demonstrated the ability to meet specified tolerances for photopolymer coating of solid microneedle arrays.

Background

Solid microneedle arrays are increasingly used in various biomedical applications including drug delivery, biosensing, and tissue engineering. A vital component of the microneedle fabrication process involves applying a photopolymer insulation layer along the height of each microneedle that covers most of the needle, leaving only a small portion of the tip uninsulated for functional interaction with tissue.

The applied insulation layer is essential to microneedle performance since it improves durability and enables precise control of the active region. However, conventional coating processes have limited control over insulation height, preventing consistent manufacturing of these devices. This project addresses the need for an automated solution that produces repeatable and reliable insulation coating results.

The dispensing method employs pneumatic actuation via a picopump system, which provides precise droplet-by-droplet deposition of photopolymer (SU-8) around each microneedle. This approach enables superior control over the micro-volumes required for coating individual microneedles. A diagram representing the concept behind SU-8 and PDMS dispensing are shown below in Figures 1 and 2.

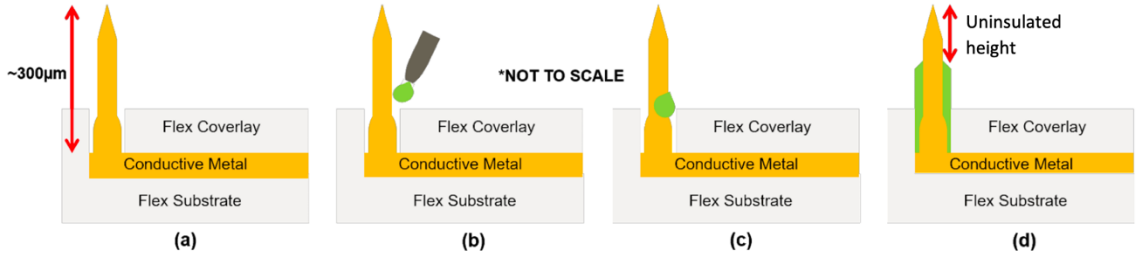


***Figure 1**: Concept Behind SU-8 Application*

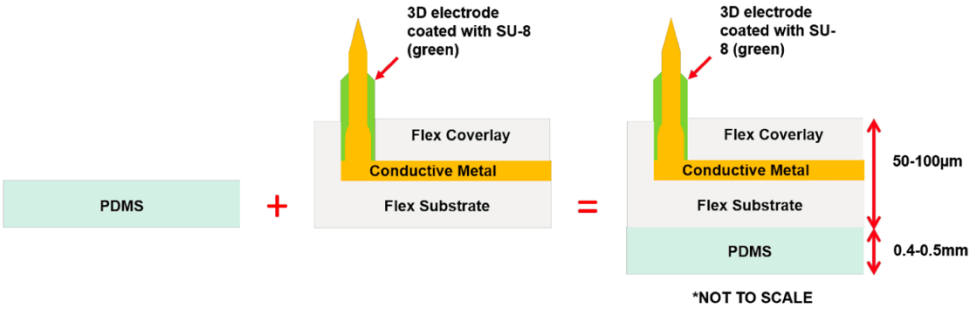


***Figure 2**: Concept Behind PDMS Application*

Problem Statement

The Dalton Lab requires an automated solution for dispensing photopolymer (SU-8) to coat solid microneedle arrays in a repeatable and reliable manner, with precise control over insulation height.

Objective

The objective of this project was to design and produce a working prototype of an automated dispenser to improve the solid microneedle array insulation process. The system functions similarly to a 3D printer, featuring a dispensing head moveable on 3 axes, attached to a frame, with a base that holds the workpiece. The development was completed in four main parts:

1. A pneumatic picopump system mounted onto an XYZ positioning system and programmed to dispense set volumes of SU-8 photopolymer via droplet-by-droplet dispensing.
2. An automated XYZ positioning system to maneuver the dispensing head around solid microneedle arrays for precise photopolymer coating.
3. A standardized process for loading the photopolymer into the dispensing system in a way that minimizes air pockets, minimizes hazard and risk to the operator, and minimizes fluid exposure to the air to prevent premature curing.
4. A workpiece fixture that can accommodate different device geometries.

Acceptance Criteria

The acceptance criteria shown in Table 1 outlines the requirements for this project, with Required, Improved, and Ideal solution tiers. Green boxes indicate criteria that were met.

Table 1: Product Acceptance Criteria

Specification	Required Solution	Improved Solution	Ideal Solution
Dispensing head travel resolution	Precision of 50µm	Precision of 20µm	Precision of 10µm
Fluid handling	System and procedure for loading photopolymer	See Required	See Required
Dispensed fluid volumes	Dispense photopolymer to coat microneedles per tolerances	Feedback loops for improved tolerances	See Improved
Insulation height tolerance	20-150µm of tip uninsulated	20-90µm of tip uninsulated	20-50µm of tip uninsulated
Optical interface	-	Small optical microscope	Digital microscope with real-time monitoring
Workpiece fixture	Different fixtures for different devices	Universal fixture for various devices	See Improved
User interface	Customizable profiles, physical controls	GUI for programming	See Improved
Safety features	Carbon filter fans, emergency stop	Advanced warnings and detection	See Improved
Documentation	SOPs for calibration, operation, maintenance	See Required	See Required

A description of each criteria is provided below:

- **Dispensing head travel resolution:** Minimum step size on the XYZ axes. This is crucial for navigating the dispensing head between the microscopic features on the microneedle arrays.
- **Fluid handling:** Develop process for loading SU-8 photopolymer into the system to be dispensed.
- **Dispensed fluid volumes:** The system must be able to adequately control dispensed volumes to consistently meet the height tolerances required.
- **Insulation height tolerance:** The microneedles should be left uninsulated between 20-150µm of their overall height at minimum (i.e., 20-150µm of the tip is left exposed for functional interaction).
- **Optical interface:** The system needs a microscope to visualize the microneedle arrays for initial setup and calibration before automatic operation.
- **Workpiece fixture:** The workpiece fixture must be able to support devices while accommodating at least 5×5 cm rectangular substrates.
- **User interface:** The system must offer automated dispensing profiles while also allowing the user to exert manual control over dispensing head position and quantity of fluid dispensed.
- **Safety features:** The system must offer basic safety features needed to safely operate with the hazardous photopolymer. These include a carbon filter fan, an emergency stop button, and appropriate SOPs.

- **Documentation:** All necessary documents required to successfully replicate and operate the device, including standard operating procedures, assembly manual, maintenance guides, and calibration information.

Design & Specifications

To meet the product acceptance criteria, the following design was created. The system features:

(1) A pneumatic picopump dispensing system mounted onto an XYZ positioning system for precise droplet dispensing;
(2) An XYZ positioning system for guiding the dispensing head to the correct location;
(3) A microscope camera for live feed of the process;
(4) A 3.5″ LCD touchscreen for GUI display;
(5) Universal workpiece fixture to accommodate various substrates;
(6) Safety features including emergency stop and carbon filter fan.

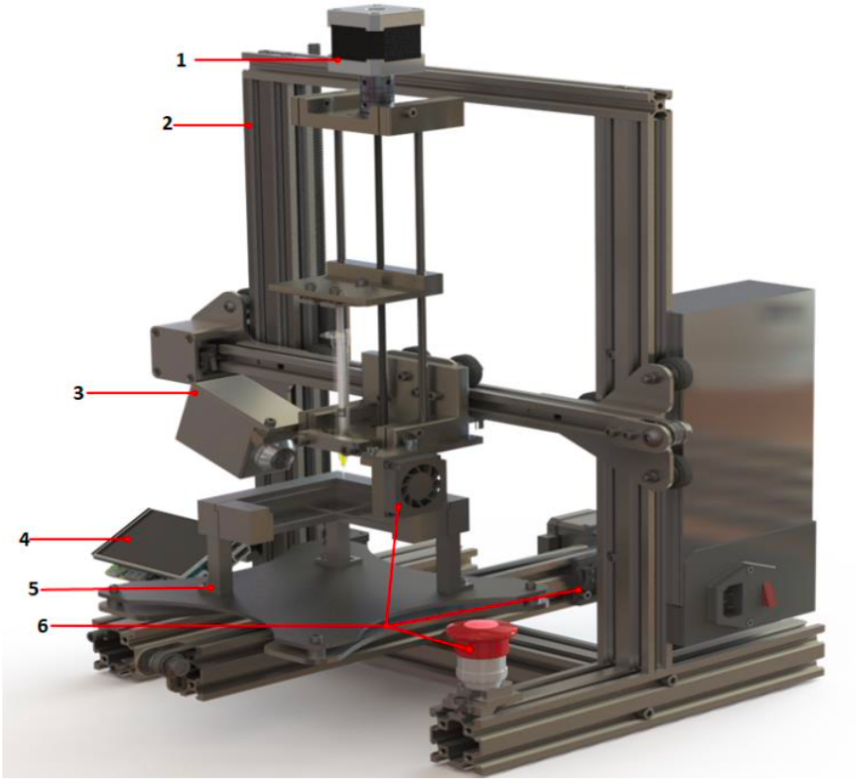


Figure 3: *CAD render of the product design*

Positioning System (Frame)

The positioning system comprises a frame (44cm × 41cm × 47cm), mechanical end stop limit switches, and 3 stepper motors. The frame was created by utilizing a commercially available 3D printer and extracting its support rails and mechanical components. One motor enables workpiece fixture movement along the Y-axis, with the remaining two motors facilitating dispensing head movement along the X and Z-axis. Movement in the X and Y axis relies on timing belts and offers a step resolution of 12.5µm based on the selected gear and belt pitch. The Z-axis is driven by a lead screw providing a travel resolution of 2.5µm.

Electrical Components

The system relies on various electronics to ensure function and support communication among components:

1. **Raspberry Pi 3B+:** Single board computer that interfaces with the Arduino over serial communication and runs the graphical user interface developed in Python.
2. **Arduino Mega 2560:** Microcontroller that controls the stepper motors and reads input from limit switches.
3. **CNC Shield:** Modular Arduino attachment that facilitates connection of stepper motor driver boards and limit switches.
4. **TMC 2208 Motor Driver:** Interfaces with motors and the control circuit to operate the motors.
5. **Camera Module (IMX477R):** Enables live view or photographs of the microneedle array being processed.
6. **Encoder:** Provides feedback signal for motor position and speed control.
7. **Power Supply:** 120VAC power supply outputting 360W at 24VDC to power all system components.
8. **Buck Converter:** DC to DC converter to step down the 24VDC supply voltage for the Raspberry Pi and carbon filter fan.

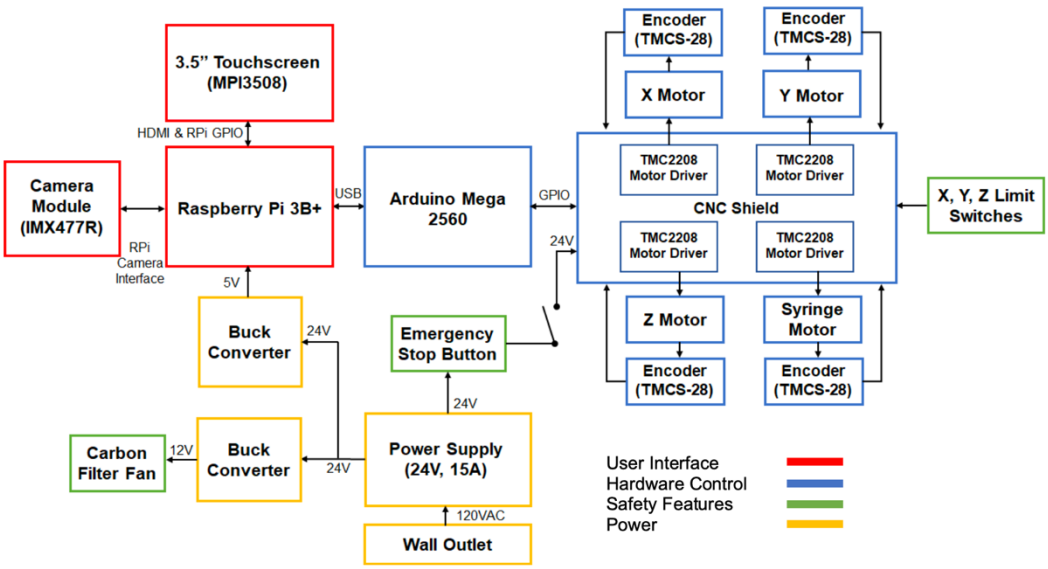


***Figure 4**: Block Diagram of the Overall System*

Dispensing Head

The designed dispensing head consists of a pneumatic picopump system with a luer lock syringe and luer lock glass capillaries for photopolymer dispensing. The picopump provides precise pneumatic actuation for droplet-by-droplet dispensing of SU-8 photopolymer. This method was selected as it performs optimally when dealing with the micro-volumes required for coating individual microneedles. The system records the dispensing parameters which, in conjunction with the known capillary dimensions, allows monitoring of the quantity of fluid ejected. The minimum dispensable volume is 1 microliter (droplet by droplet), based on the picopump pressure and pulse duration settings.



***Figure 5**: Syringe Pump Model*

Workpiece Fixture

The workpiece fixture was designed as an 11cm × 11cm clamping system capable of holding devices for photopolymer dispensing procedures. The mechanism consists of two guide rails with adjustable sliders capable of changing position along the X and Y axis to adequately grip the microneedle array substrate. This universal design allows accommodation of various device geometries while reducing manufacturing and replacement costs. The fixture sits 7cm above the Y-axis frame rail to allow proper spacing between the dispensing head attachments and the microneedle arrays.

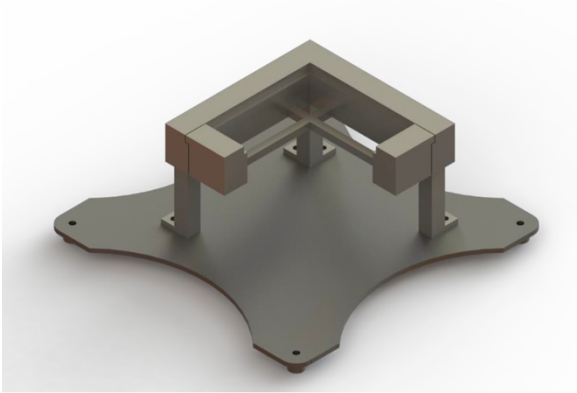


***Figure 6**: Workpiece Fixture Model*

User Interface & Programming

Python was used as the primary programming language due to its efficiency and simplicity, allowing users to generate custom profiles with relative ease. The Raspberry Pi serves as the master device while the Arduino acts as a slave device for motor control. The custom GUI allows interaction through any monitor connected to the Raspberry Pi. The Arduino executes dedicated tasks such as motor operations while the Raspberry Pi serves as the main computer interface. Manual features include the ability to move the dispensing head in all 3 dimensions, set custom travel resolution for each axis, and dispense working fluid on command.

Results

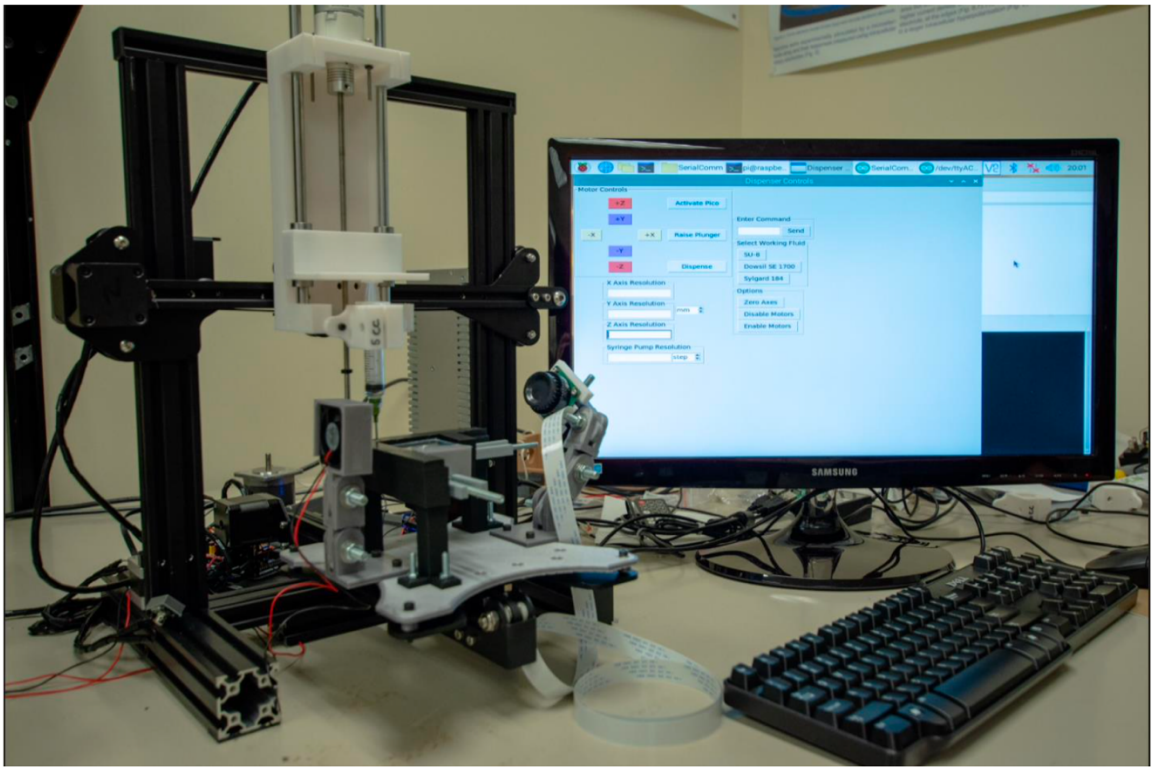


***Figure 7**: Assembled Dispensing System*

Dispensing Head Travel Resolution

The dispensing head achieved a travel resolution of 12.5µm in the X-axis and Y-axis, and 2.5µm in the Z-axis. XYZ step resolution was measured and calibrated using a microscope micrometer ruler calibration slide, commanding the system to move a specified distance, and adjusting the steps-per-millimeter constant in the Arduino code until commanded steps equaled actual steps. This places the travel resolutions into the improved solution criteria for X/Y and ideal criteria for Z-axis.

Fluid Handling

The fluid handling procedure evolved during development. Initially, a reservoir system was considered; however, due to the curing process of SU-8 when exposed to air, heat, or light, smaller volumes extracted from the stock container at dispensing time were implemented. Due to the small volumes needed, a 50µm × 5cm microcapillary is sufficient for at least 3-4 microneedle array devices with 60 needles each. The exact number of devices that can be coated depends on the dimensions of the workpiece. This design meets the required solution.

Dispensed Fluid Volumes

Photopolymer dispensing is achieved using a pneumatic picopump system for droplet-by-droplet deposition. This method was found during testing to perform optimally when dealing with the micro-volumes required for the coating process. The pump is actuated through the Arduino/Raspberry Pi interface, and capillaries are mounted using the existing design. This method meets the required solutions for precise micro-volume dispensing.

Photopolymer Insulation Height Tolerance

SU-8 photopolymer was successfully coated on an 8×8 solid microneedle array with 300µm needle spacing (n=1), where each microneedle was 300µm tall. Calibration involved dispensing droplets next to microfeatures with known size to determine the relative size of the droplet. Due to limited availability of test workpieces, the sample size could not be increased. The required criteria of 20-150µm of the microneedle tip being uninsulated was confirmed qualitatively by comparing the SU-8 height with the known height of the microneedles.

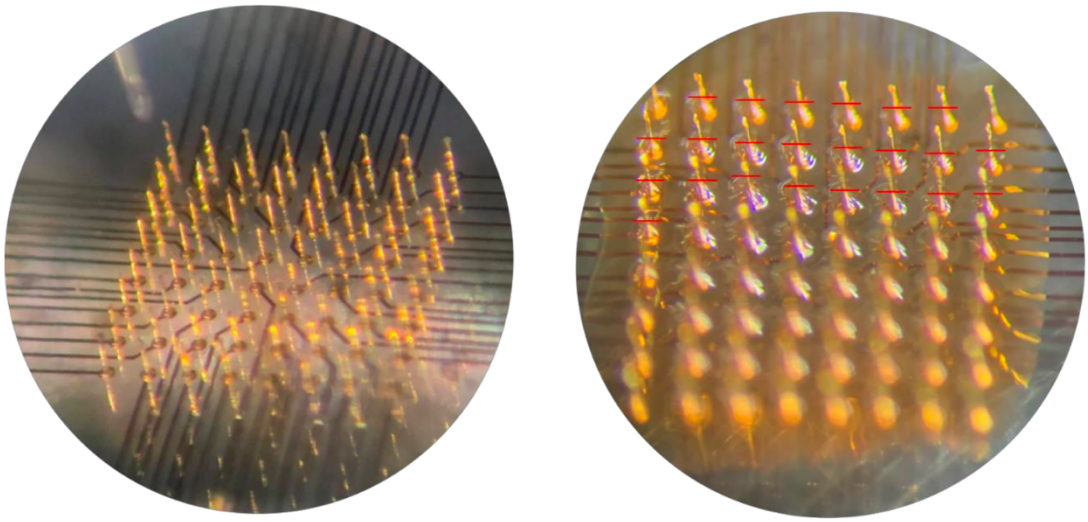


***Figure 8**: Left shows 3D microelectrodes prior to coating, right shows after.*

Optical Interface

The improved solution was achieved where the system can be used with a standalone desktop optical microscope to track dispensing head movement and observe microfeatures on the finished workpiece. An articulating mount was designed and manufactured for a digital microscope where the digital output can be integrated with the custom GUI through Python libraries. The digital microscope was connected and tested but requires enhanced shock absorption to dampen vibrations from system operation.

Workpiece Fixture

The final universal workpiece fixture fits all provided workpieces and meets the improved requirements. The design offers universal accessibility, saving development time that would otherwise be spent creating individual designs for various devices.

User Interface

The system is operated using a monitor connected to the Raspberry Pi rather than the originally planned 3.5" LCD screen. This change was made because the small screen proved cumbersome to operate and required constant cleaning. Mouse/keyboard control was found to be easier and cleaner. The required solution is met with manual control buttons and one-click profiles for device dimensions. User customization is available through code modification, which is straightforward for users with basic programming experience.

Safety Features

The required solution was met with basic safety features including: (1) a carbon filter fan to absorb material fumes, and (2) two emergency stop methods. The first is a software emergency stop triggered by pressing any button while the code runs. The second is a hardware emergency stop button that disconnects power supply from the motors when pushed and reconnects when twisted.

Documentation

All necessary documentation was provided meeting the required solution. This includes: Fluid Loading Procedure, Raspberry Pi and Arduino Code, CAD Files and Schematics, and System Operations and Assembly Manual.

Product Specifications

Table 2: Mechanical Specifications

Feature	Specification
Overall Dimensions	44cm × 41cm × 47cm
Dispensing Head Travel Resolution	X & Y: 12.5µm; Z: 2.5µm
Workpiece Fixture	Universal (supports substrates up to 5×5cm)

Table 3: Electrical Specifications

Feature	Specification
Power	360W at 24V 15A DC switching power supply
Motor Drivers	TMC2208
Motor Controller	Arduino MEGA2560
Motor Connector Board	Arduino CNC Shield
Interface Computer	Raspberry Pi 3B+
Motors	1.8° NEMA 17 biphasic stepper motors

Table 4: Microscope Specifications

Feature	Specification
Image Sensor	Sony IMX477
Objective Lens	Compatible with C and CS mount lenses
Data Interface	Raspberry Pi camera port

Table 5: Dispensing System Specifications

Feature	Specification
Dispensing Method	Pneumatic picopump (droplet-by-droplet)
Minimum Dispensable Volume	1µL
Dispensing Tip	Luer lock glass capillaries (30-70µm diameter)

Future Improvements

The following improvements are proposed for further development:

Hardware

Materials and Construction

Consider replacing 3D printed parts with metal for improved durability and dimensional accuracy. Some shrinkage occurs during printing that causes minor fit issues. Machining parts such as lead screws and smooth rods to correct dimensions would improve precision and accuracy.

Workpiece Fixture Range

The current fixture is compatible with substrates from approximately 30mm to 80mm. A metal fixture would allow thinner supports and accommodation of smaller sizes. A modular fixture with interchangeable pieces could accommodate a wider range of device sizes.

Implement Encoders

The current setup uses feedforward control without fully implemented encoders. Implementing encoders would enable closed-loop control with feedback (e.g., PID control), improving system accuracy and enabling more advanced control logic.

Microscope Support

The current microscope mounting causes noticeable vibration during operation. This could be addressed by attaching the stand directly to the frame, implementing vibration dampers, or replacing 3D printed stand components with metal ones. Installing a long working distance microscope objective would also reduce collision risk.

Software

Code Optimization

The code structure should be modified to adjust dispensing parameters based on fluid properties as user inputs. The current 9600 baud rate for serial communication causes some lag; increasing baud rate currently causes reliability issues. Consider writing custom stepper motor control code to replace the AccelStepper library which has minor bugs.

Home Position Feature

End stop switches are installed but not fully implemented. Completing the zeroing feature would save time repositioning the dispensing head and improve result consistency.

Collision Avoidance

Code and hardware (e.g., end stop switches, ultrasonic sensors) should be implemented to prevent the dispensing head from hitting the microscope during operation.

Complex Shapes

Currently, the code only allows straight-line movements. Implementing arc moves or higher-level functions to approximate curves with small straight moves would be beneficial for non-rectangular substrate profiles.

Dispensing

Insulation Height Measurement

Using a calibrated microscope, insulation heights can be measured quantitatively. This requires developing a custom microscope adaptor for microneedle arrays that allows both the needle and insulation to be in focus. Alternatively, a flying probe tester could measure impedance before and after coating to correlate with insulation height.

Improve Capillary Luer-Lock Adapters

Air bubbles sometimes form in capillaries during dispensing, affecting consistency and accuracy. The current in-house adapters (glass capillary joined to luer lock cap with adhesive) should be replaced with off-the-shelf luer lock capillary adapters (available from World Precision Instruments) to prevent air bubble formation.

Deliverables

The following deliverables are provided with this system:

- Fluid Loading Procedure documentation
- Raspberry Pi and Arduino source code
- CAD files (STL format) and schematics
- System Operations and Assembly Manual
- Calibration and maintenance guides